

"Real vs simulated: what actually ran"

Which parts of the C1 POC exercised the real mechanism, which are faithful local substitutes, what was recorded, what was injected, and what is projected or argued rather than measured.

C1

C1 · Production AI Engineering

2 min read · Published 2026-10-08 · Real vs simulated

Real

the actual mechanism was exercised.

- **Model inference:** `gpt-oss:20b` served locally by Ollama, for every reasoning component of every architecture (72 blind workflows, plus E6 and E8).
- **MCP:** four MCP stdio servers on the official Python SDK (`mcp==2.2.0`), started as separate processes by the host and by each agent process.
- **A2A:** agents as separate OS processes serving A2A v1.0 over JSON-RPC with the official `a2a-sdk 1.2.2`; Agent Card discovery at `/.well-known/agent-card.json` ; `SendStreamingMessage` for every delegation (task lifecycle observed on the wire); `GetTask` and `CancelTask` .
- **Authentication across the boundary:** every A2A request carries a token in `Authorization: Bearer` , verified by the agent before any work (signature, audience, expiry, workflow, delegation).
- **Process failure:** SIGKILL of a live agent process (E6), restart by the supervisor, `GetTask` against the restarted server.
- **Retries, timeouts, idempotency keys** honoured by the system of record.
- **OpenTelemetry:** one trace per incident across host and agent processes via W3C `traceparent` .
- **Measurements:** token counts reported by the model server; wall-clock latency on one machine; bytes on the wire for every A2A delegation.

Simulated

a faithful local substitute for an external system.

- checkout-api and every other enterprise system (incidents, deployments, configuration, flags, metrics, logs, runbooks), in one SQLite world with a simulated clock.
- Every remediation (rollbacks, restarts, scaling, config reverts, flag changes) and its effect on service health.
- The token service (HMAC-signed tokens with RFC 8693-style actor chains; not OAuth 2.0).
- The incident commander's approval (a simulated approver that approves whatever policy routes to approval, bound to a digest of the exact call).
- The external card-network provider and its status page (B3).

Recorded

captured model traffic, reused for deterministic re-execution.

- Every model call of the blind run on an HTTP tape (`coordination_poc/runs/2026-10-08-blind/tape/`); `coord.verify_replay` re-executes the run from the tape with no model, across all processes, and compares outcomes and ledgers.

Injected

a failure introduced on purpose.

- E6: a SIGKILL of the diagnosis agent mid-task (K1) and of the remediation agent after its write (K2, K2-neg).
- E8: removal of the termination owner (no hop limit, no cycle check, a budget three times larger; safety cap only).
- In the fixtures: distractor releases, coincident changes, an unsafe suggestion in an incident note (B7), an irreversible migration (B6), an already-recovered alert (B8).

Scripted

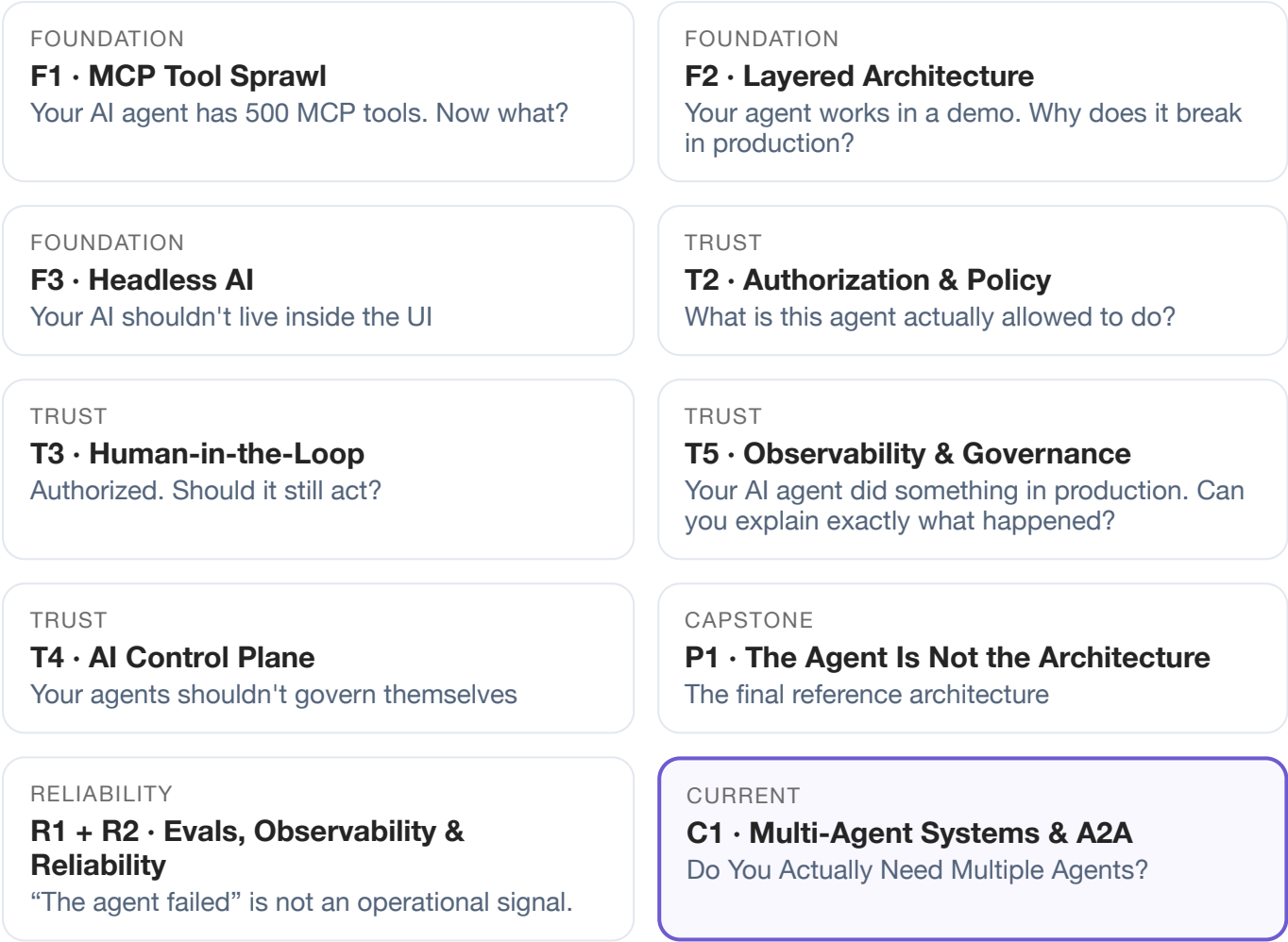
no model; used where removing model variance is the point.

- E7 (A2A boundary cost) and the 64 tests use a deterministic stand-in model (`coord/scripted.py`). No headline number comes from it.

Not measured

- Dollar cost: local inference has no API price; the article reports tokens only.
- Behaviour at production concurrency, TLS, multi-tenant agents, other models or other agent frameworks.
- Human approval latency (approval is synchronous and simulated).

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The full learning map (Start Here)

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Production AI Engineering · C1. Every measured number here is substituted at build time from `coordination_poc/runs/2026-10-08-blind/facts.json`. Figures carry their provenance.

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